

SHAPING OF THE EARTH'S SURFACE

Landforms & Plate Tectonics

1. Landforms

- **Landforms** are natural features found on the Earth's surface.
- They are formed by processes like **weathering, erosion, deposition, and movement of the Earth's crust.**
- Examples include **mountains, volcanoes, plains, valleys, plateaus, deserts, and coastal features.**

2. Plate Tectonics

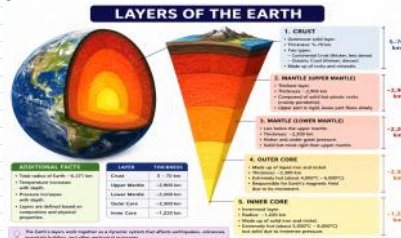
- **Plate tectonics** is a theory given by **W. J. Morgan** that explains the movement of the Earth's crust.
- The Earth's outermost layer is divided into several **large and small tectonic plates.**
- These plates move slowly over the **semi-molten layer beneath them.**
- Plate movement causes major features and natural phenomena such as **mountains, earthquakes, and volcanoes.**
- It also helps explain the **formation of continents and oceans.**

Earth's Interior & Plate Tectonics

1. Layers of the Earth

The Earth has three main layers:

- **Crust** – The outermost layer on which we live.
- **Mantle** – A very thick and hot layer below the crust.
- **Core** – The innermost, extremely hot and heavy layer.



2. Lithosphere

- The **lithosphere** consists of the crust and the upper part of the mantle.
- It is divided into several **tectonic plates.**
- The lithosphere is about **100 km thick.**

3. Asthenosphere

- The **asthenosphere** lies beneath the lithosphere.
- It is a **hot, mobile layer of partially molten rock.**
- It allows tectonic plates to move.

4. Core

- **Outer Core:** A fluid layer mainly made of iron and nickel.
- **Inner Core:** A solid, extremely hot and dense part of the Earth.

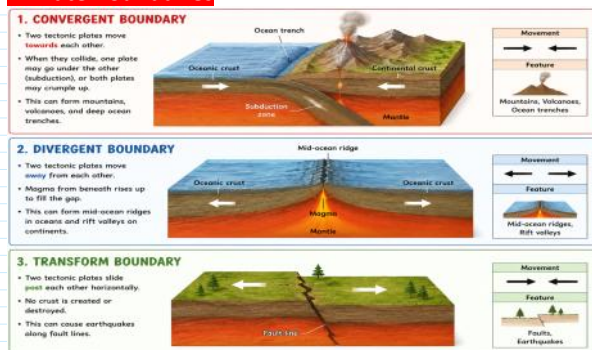
5. Tectonic Plates

- Tectonic plates are **large slabs of solid rock** that move very slowly.
- They generally move a few **centimetres per year.**
- Three main types are:
 - **Continental plates** – carry continents.
 - **Oceanic plates** – carry ocean floors.
 - **Mixed plates** – carry both continents and oceans.
- Major plates include the **Pacific, Eurasian, African, North American, South American, Indo-Australian, and Antarctic Plates.**

6. Convection Currents

- The movement of tectonic plates is mainly caused by **convection currents in the mantle.**
- Heat from the Earth's core makes hot material **rise**, while cooler material **sinks.**
- This continuous movement creates convection currents that **push and pull tectonic plates.**

7. Plate Boundaries



Convergent Boundary

- Two plates move **towards each other.**
- Continental plates may form **fold mountains**, such as the Himalayas.
- An oceanic plate colliding with a continental plate can cause **volcanic activity and earthquakes.**

Divergent Boundary

- Two plates move **away from each other.**
- Magma rises and forms **new crust.**
- Example: **Mid-Atlantic Ridge.**

Transform Boundary

- Two plates **slide past each other**.
- Crust is neither created nor destroyed.
- This movement mainly causes **earthquakes**.
- Example: **San Andreas Fault** in the United States.

Plate Tectonics, Earthquakes & Volcanoes

1. Importance of Plate Tectonics

- Plate tectonics plays an important role in **shaping the Earth's surface**.
- Plate movement forms **mountains, valleys, ocean basins, volcanoes, and earthquakes**.
- It explains the distribution of **continents and oceans**.
- Most earthquakes and volcanoes occur along **plate boundaries**.

2. Ring of Fire



- The **Ring of Fire** is an area around the **Pacific Ocean**.
- It has a high concentration of **earthquakes and volcanoes**.
- This is mainly due to the movement and interaction of tectonic plates.

3. Earthquakes in India

- India has experienced several **major earthquakes** in the past.
- Earthquakes can cause serious damage to **human life, buildings, and the environment**.
- Densely populated areas are especially vulnerable to earthquake damage.

4. Earthquakes in Ancient India

- In ancient times, earthquakes were known as **"Bhukampa"**, meaning the shaking of the Earth.
- **Varahamihira** discussed earthquakes in the *Brihatsamhita*.
- He connected earthquakes with observations of **wind, rain, clouds, animal behaviour, and planetary alignments**.

5. Mud Volcano

- The **Mud Volcano at Baratang Island** is a rare natural feature of India.
- It is located in the **Andaman and Nicobar Islands**.
- Unlike a fiery volcano, it produces **mud** due to natural underground gases and pressure.

5. Weathering

Weathering is the process by which rocks on the Earth's surface **break down into smaller pieces**.

Important: Weathering does **not involve the movement** of broken material.

Types of Weathering

A. Physical Weathering

Rocks break into smaller pieces without changing their chemical composition.

Causes:

- Heating and cooling
- Water
- Wind
- Frost/ice expansion

B. Chemical Weathering

Minerals in rocks change because of chemical reactions with:

- Water
- Air
- Acids

These reactions may produce new substances.

C. Biological Weathering

Weathering caused by:

- Plants
- Animals
- Micro-organisms

Example: Plant roots grow into cracks in rocks and split them apart.

Importance of Weathering

- Helps shape the Earth's surface.
- Helps in the formation of **soil**.

6. Erosion

Erosion is the process by which soil, rocks and other surface materials are **worn away and carried from one place to another** by natural agents.

Weathering vs Erosion

Weathering

Breaks rocks down
No movement of material
Occurs at the place where rocks break down

Erosion

Wears away and carries materials
Involves movement of material
Material is transported to another place

Types of Erosion

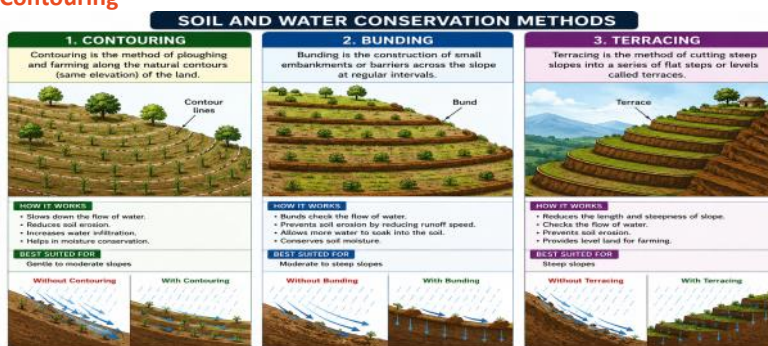
1. **Water Erosion** – caused by rivers, rain and ocean waves.
2. **Wind Erosion** – common in dry and sandy areas.
3. **Glacial Erosion** – caused by moving ice/glaciers.
4. **Coastal Erosion** – caused by sea waves wearing away coastal land.

Effects of Erosion on Humans

- Removes fertile topsoil.
- Reduces crop yields.
- Can damage houses and roads.
- Can wash away land near rivers and coasts.
- Creates problems for construction and mining.
- Can affect tourism and fishing.

7. Soil and Water Conservation Methods

Contouring



A water-conservation technique in which trenches are dug along the **contour lines of a hillside**.

Purpose:

- Slow down water runoff.
- Increase water infiltration.
- Reduce soil erosion.
- Recharge groundwater.

Bunding

Building **earthen embankments** along contour lines.

Purpose:

- Slow surface runoff.
- Reduce soil erosion.
- Increase water infiltration.
- Improve soil moisture.

Terracing

Creating a series of **level or gently sloping steps** on a hillside.

Purpose:

- Slow down water flow.
- Prevent soil erosion.
- Help conserve soil and water.

Zabo System

- A traditional farming and water-management system found in **Nagaland**.
- It uses earthen bunds on hillslopes.
- It helps in **soil and water conservation**.

Check Dams

- Small dams built across streams.
- Reduce water velocity.
- Prevent soil erosion.
- Allow sediment deposition.
- Help in groundwater recharge.

Agents of Gradation

Agents of gradation are natural forces that **erode, transport, and deposit materials** on the Earth's surface. They gradually help to **level or smooth the Earth's surface**.

Main Agents of Gradation

1. Running Water

- Erodes rocks and soil.
- Transports materials.
- Helps form **valleys and plains**.

2. Glaciers

- Scrape and carry large amounts of rock and soil.
- Form **U-shaped valleys**.

3. Wind

- Erodes and transports sand.
- Deposits sand and helps shape **desert landforms**.

4. Sea Waves

- Erode coastal rocks.
- Form **cliffs, beaches, and bays**.

5. Groundwater

- Dissolves rocks such as **limestone**.
- Forms **caves and sinkholes**.

Overall Role

Together, these agents continuously modify the Earth's landforms by:

- **Lowering high areas**
- **Filling low areas**
- **Eroding and transporting materials**
- **Depositing materials**
- Gradually making the Earth's surface more **level and smooth**.

Landforms and Human Civilisation

1. Role of Landforms in Human History

Landforms have played an important role in the development of **human civilisations**.

2. Rivers and Fertile Plains

- Rivers such as the **Ganga, Nile, Brahmaputra, and Indus** supported agriculture.
- Fertile plains helped in the development of **early cities and agricultural societies**.

3. Mountains

- Mountains acted as **natural barriers and protectors**.
- The **Himalayas** helped protect India from invasions.
- Mountain passes, such as the **Khyber Pass**, also allowed cultural exchange and movement.

4. Deserts

- Deserts such as the **Thar Desert** limited the development of large settlements.
- However, deserts also encouraged the development of **trade routes**, such as the **Silk Route**.

5. Coasts and Harbours

- Coasts and natural harbours supported:
 - Trade
 - Travel
 - Cultural contacts
- They helped kingdoms, especially in **South India**, develop and prosper.

6. Overall Importance

Physical features of the land have influenced:

- **Wars**
- **Settlements**
- **Agriculture**
- **Trade**
- **Cultural exchange**
- **Economic growth**

Conclusion: Landforms have strongly influenced the history, settlement patterns, trade, and cultural development of human societies.

Running Water – Landforms

1. Running Water

Rivers shape the land through three main processes:

- **Erosion** – removal of soil and rocks.
- **Transportation** – carrying sediments from one place to another.
- **Deposition** – dropping sediments when the river loses energy.

Landforms in Different River Courses

Upper Course:

- V-shaped valleys
- Waterfalls
- Rapids

Middle Course:

- Meanders
- Oxbow lakes
- Floodplains

Lower Course:

- Deltas
- Levees
- Alluvial fans

2. Waterfall

A **waterfall** is a landform where a river flows over a steep cliff or vertical drop.

Formation

- Waterfalls usually form in the **upper course** of a river.
- Hard rock resists erosion, while softer rock is eroded faster.
- This creates a sudden drop in the river channel.
- A **plunge pool** may form at the base of the waterfall.

Importance

- Attractive natural features and tourist destinations.
- Support local tourism and the economy.
- Can be used for **hydroelectric power generation**.
- Provide opportunities for trekking, photography and recreation.
- Some waterfalls have cultural or religious importance.

3. Meander

A **meander** is a winding curve or bend in the middle or lower course of a river.

Formation

- The river erodes the **outer bank** of a bend.
- It deposits sediments on the **inner bank**.
- Continuous erosion and deposition gradually create large loops.

Importance

- Deposited sediments create **fertile soil**.
- Suitable areas for agriculture.
- Villages and towns may develop near meanders.
- Meandering rivers can support **navigation and irrigation**.
- They may also attract tourists because of their scenic landscapes.

Example: The Grand Anicut (Kallanai) in Tamil Nadu is an example of river use for irrigation.

4. Delta

A **delta** is a landform formed at the **mouth of a river**, where it enters a sea, ocean or lake.

Formation

- Rivers carry sediments from upstream.
- When the river enters a sea, ocean or lake, its speed decreases.
- The river deposits its sediments.
- Over time, the deposits form a **fan-shaped or triangular area of land**.

Importance

- Deltas have very **fertile alluvial soil**.
- Suitable for agriculture.
- Support crops such as **rice and jute**.
- Important for fishing because freshwater mixes with seawater.
- Support dense human settlements.
- Rivers provide routes for trade and transportation.

Problem

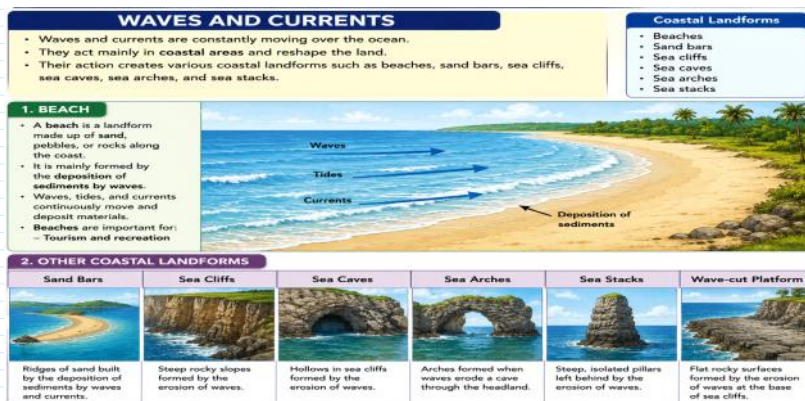
- Deltas are often **prone to flooding**, which can affect human life and activities.

Quick Revision

Landform	Main Process	Common Location
Waterfall	Erosion	Upper course
Meander	Erosion + Deposition	Middle/Lower course
Delta	Deposition	River mouth

Waves and Currents –

1. Waves and Currents



- **Waves and currents** are constantly moving over the ocean surface.
- They act mainly in **coastal areas** and reshape the land along the **coastal zone**.
- Their action creates various coastal landforms such as:
 - Beaches
 - Sand bars
 - Sea cliffs
 - Sea caves
 - Sea arches
 - Sea stacks

2. Beach

- A **beach** is a landform made up of **sand, pebbles, or rocks** along the shoreline of a sea, ocean, or lake.
- It is mainly formed by the **deposition of sediments by waves**.
- Waves, tides, and currents continuously move and deposit materials along the coast.
- Beaches are important for:
 - **Tourism and recreation**
 - Swimming and relaxation
 - Fishing
 - Collection of sand and shells
- Beaches can also act as **natural barriers** against strong waves and coastal erosion, helping protect coastal settlements.

3. Coastal Erosion

- **Coastal erosion** occurs when **waves, tides, and currents wear away land** along the coast.
- Continuous erosion creates several distinctive coastal landforms.

Major Landforms of Coastal Erosion

Landform

Sea Cliff

Wave-cut Platform

Sea Cave

Sea Arch

Sea Stack

Formation

A steep rock face formed when waves erode and undercut the base of the coast.

A relatively flat area left behind as a sea cliff gradually retreats.

Formed when waves erode weak parts of coastal rocks.

Forms when erosion enlarges a cave or caves on opposite sides of a headland meet.

An isolated pillar of rock left standing after a sea arch collapses.

4. Sequence of Formation

Sea Cave → Sea Arch → Sea Stack

Continuous wave erosion weakens the rock → the cave becomes larger → an arch develops → the arch eventually collapses → a **sea stack** remains.

Importance

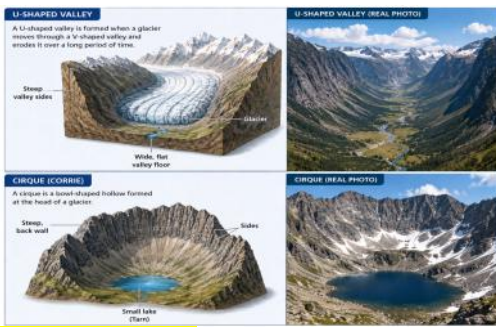
- Coastal landforms shape the **coastal landscape**.
- Some coastal areas attract tourists.
- Coastal regions may require **protection** to safeguard human settlements from erosion.

Glacial Erosion

Glacial erosion occurs when glaciers slowly move over the land and **erode, carve and reshape the Earth's surface**.

A glacier is a large mass of ice that moves slowly under its own weight. While moving, it carries rocks, soil and other debris, which help in eroding the land.

Major Landforms of Glacial Erosion



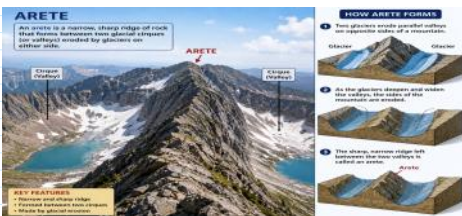
1. U-shaped Valley

- A **U-shaped valley** is formed when a glacier widens and deepens an existing river valley.
- It has a broad floor and steep sides.
- River valleys are generally V-shaped, while glacial valleys are usually U-shaped.
- These valleys are important for **tourism, trekking and agriculture**.

2. Cirque

- A **cirque** is a **bowl-shaped depression** found at the head or upper part of a glacier.
- It is formed mainly by glacial erosion.
- After the glacier melts, a small lake may sometimes develop in the depression.

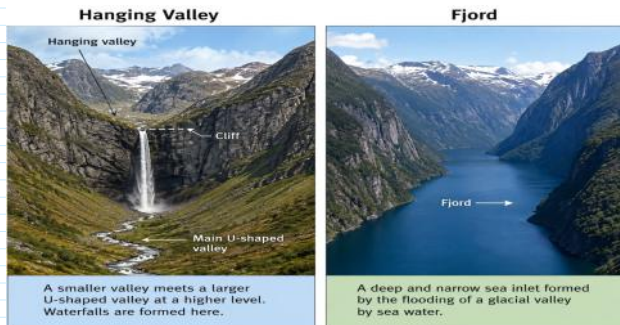
3. Arete



An **arete** is a **sharp, narrow ridge** between two glacial valleys or cirques.

- It develops when glaciers erode a mountain from opposite sides.
- Continued erosion makes the ridge steep and sharp.

4. Hanging Valley



A **hanging valley** is a smaller glacial valley that joins a deeper main valley at a higher level.

- It forms when a smaller glacier joins a larger glacier.
- After the ice melts, waterfalls may develop at the edge of the hanging valley.

5. Fjord

- A **fjord** is a **deep, narrow sea inlet surrounded by steep sides**.
- It forms when a glacier cuts a deep valley and the valley is later flooded by seawater.
- Fjords are useful for **harbours, fishing and tourism**.

Importance of Glaciers and Glacial Landforms

- Glaciers are important sources of **fresh water**.
- Meltwater from glaciers feeds rivers and supports people living downstream.
- U-shaped valleys and cirques attract tourists for **trekking, skiing and mountaineering**.
- Fjords are useful for **harbours and fishing**.
- Fertile soils in some glacial valleys support **agriculture**.

Moraines

Moraines are landforms formed by the **deposition of rocks, soil and debris (called till)** carried by glaciers. When a glacier melts or retreats, the materials carried by it are deposited on the land and form moraines.

Types of Moraines

1. Lateral Moraine

- Forms along the **sides of a glacier**.
- It consists mainly of rocks and debris deposited along the glacier's edges.

2. Terminal Moraine

- Forms at the **end (terminus)** of a glacier.
- It marks the **farthest point reached by the glacier** during its advance.

3. Medial Moraine

- Forms in the **middle** of a glacier.
- It develops when **two glaciers meet** and their lateral moraines join together.



Importance of Moraines

- They can create **fertile soil** for agriculture.
- They can form **natural dams and lakes**.
- Such lakes may provide water for **water supply and irrigation**.
- In some cases, water resources associated with them can be used for **hydroelectric power**.

4. Wind Erosion

Wind erosion occurs when strong winds pick up and carry away loose particles of **sand, soil and dust**, gradually changing the landscape. Wind erosion is especially important in **desert and arid regions**.

Major Landforms of Wind Erosion



1. Yardangs

- **Yardangs** are streamlined ridges of rock carved by wind erosion.
- Wind removes softer rock faster than harder rock.
- They are commonly found in dry desert regions.

2. Ventifacts

- **Ventifacts** are rocks that have been **polished, scratched and shaped by wind-blown sand**.
- Continuous sandblasting changes their surfaces and shapes.

3. Blowouts / Wind Hollows

- **Blowouts** are shallow depressions formed when wind removes loose sand and other materials.
- They are common in areas containing loose and dry sediments.

4. Desert Pavements

- **Desert pavements** are flat surfaces covered mainly with **stones and pebbles**.
- They form when wind removes the finer particles such as sand and dust.
- The heavier stones are left behind on the surface.

5. Sand Dunes



Sand dunes are hills or ridges of sand formed by the action of wind.

They are commonly found in:

- Deserts

- Sandy coastal areas

Wind transports sand and deposits it when its speed decreases. Continuous deposition gradually builds up dunes.

Types of Sand Dunes

1. Barchan Dunes

- **Crescent-shaped** dunes.
- Form where the supply of sand is **limited**.
- Develop mainly where wind blows predominantly from **one direction**.
- Their pointed ends are called **horns**.

2. Longitudinal Dunes

- Also called **linear dunes**.
- They are **long, narrow ridges of sand**.
- They generally form **parallel to the prevailing wind direction**.
- They can extend over long distances.

3. Star Dunes

- Star dunes have a **central peak with several arms**.
- They form where winds blow from **different directions**.
- They are generally large and complex dunes.

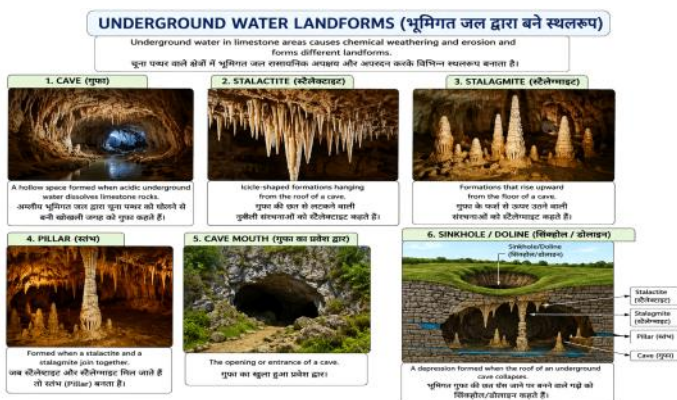
4. Parabolic Dunes

- Generally **U-shaped** dunes.
- They often form where **vegetation stabilizes part of the sand**.
- They are commonly found in coastal sandy areas and some other sandy environments.

Quick Revision Chart

Process	Landforms
Glacial Erosion	U-shaped valley, Cirque, Arete, Hanging valley, Fjord
Glacial Deposition	Moraines
Types of Moraines	Lateral, Terminal, Medial
Wind Erosion	Yardangs, Ventifacts, Blowouts/Wind hollows, Desert pavements
Wind Deposition	Sand dunes
Types of Sand Dunes	Barchan, Longitudinal, Star, Parabolic

UNDERGROUND WATER AND KARST LANDFORMS



Underground water is water stored beneath the Earth's surface in soil, rocks and fractures. In areas dominated by **limestone and other soluble rocks**, underground water plays an important role in modifying the landscape.

The action of underground water mainly occurs through **chemical weathering, dissolution and erosion**, producing a distinctive landscape known as **Karst Topography**.

Karst Topography

Karst topography develops where soluble rocks, particularly **limestone**, are dissolved by slightly acidic groundwater.

Process:

Rainwater + Carbon dioxide → weak carbonic acid → dissolution of limestone

This process gradually enlarges cracks, joints and underground cavities.

Major Landforms

1. Caves

Caves are **underground cavities** formed when acidic groundwater dissolves soluble rocks.

- Develop mainly in limestone regions.
- May contain stalactites, stalagmites and pillars.
- Some caves contain underground streams and rivers.

2. Stalactites

Stalactites are **icicle-shaped mineral deposits** hanging from the roof of a cave.

Memory Trick:

Stalactite = Ceiling → T

3. Stalagmites

Stalagmites are mineral deposits that **rise upward from the floor of a cave**.

They develop when mineral-rich water drips from the cave roof and deposits minerals on the floor.

Memory Trick:

Stalagmite = Ground → G

4. Pillars / Columns

When a stalactite and a stalagmite grow until they join, they form a **pillar or column**.

5. Sinkholes / Dolines

A **sinkhole or doline** is a depression formed when the ground collapses into an underground cavity or when soluble rock is progressively dissolved.

6. Underground Rivers

Underground rivers are water channels that flow through **caves and subterranean drainage systems**.

Importance of Underground-Water Landforms

- Provide **freshwater resources**.
- Support underground drainage systems.
- Promote **cave tourism**.
- Have geological and scientific importance.
- Some caves have **cultural and religious significance**.
- Stalactites and stalagmites help scientists understand long-term environmental changes.

Key Terms

Karst → Limestone dissolution → Caves → Stalactites → Stalagmites → Pillars → Sinkholes → Underground rivers

LANDFORMS AND DISASTERS

LANDSLIDES

A **landslide** is the rapid or gradual downslope movement of soil, rocks, debris or a combination of these materials under the influence of **gravity**.

It is a major form of **mass wasting**.

Major Causes

A. Natural Causes

1. **Heavy and prolonged rainfall**
 - Water infiltrates the soil.
 - Increases the weight of the slope.
 - Reduces friction between particles.
 - Makes the slope unstable.
2. **Earthquakes**
 - Ground vibrations weaken slopes.
 - Can trigger sudden mass movement.
3. **Volcanic activity**
 - Volcanic materials can destabilise slopes.
 - Eruptions and volcanic earthquakes may initiate landslides.
4. **Steep slopes**
 - Greater slope angle increases the downslope component of gravity.
5. **Weathering**
 - Weakens rocks and makes them more susceptible to failure.
6. **Loose or fractured rocks**
 - Provide less resistance to gravitational movement.

B. Human Causes

- **Deforestation**
- **Mining**
- **Road construction**
- **Unplanned construction on hillsides**
- **Improper land use**
- **Poor drainage systems**
- **Excessive removal of vegetation**

Role of Water

Water is one of the most important triggers of landslides.

Rainfall → Infiltration → Increased pore-water pressure → Reduced friction/shear strength → Slope failure

Effects

- Loss of human and animal lives
- Destruction of houses and infrastructure
- Blocking of roads and railway lines
- River blockage and temporary lakes
- Agricultural land loss
- Disruption of communication and transport

- Secondary flooding

Prevention and Mitigation

- Afforestation and vegetation restoration
- Proper drainage systems
- Controlled construction
- Slope stabilisation
- Retaining walls
- Terracing
- Rock bolting and slope reinforcement
- Landslide hazard mapping
- Early-warning systems

Key Concept

Slope stability depends on the balance between driving forces and resisting forces.

When **driving forces** > **resisting forces**, slope failure may occur.

AVALANCHES

An **avalanche** is the sudden downward movement of a large mass of snow, ice and sometimes rocks or debris along a mountain slope.

It occurs mainly on **steep, snow-covered slopes**.

Major Causes

1. Heavy Snowfall

Rapid accumulation of snow increases the load on the snowpack.

2. Weak Snow Layers

If new snow rests on weak or poorly bonded layers, the snowpack becomes unstable.

3. Sudden Temperature Rise

Rapid warming can partially melt snow and reduce the friction holding different layers together.

4. Strong Winds

Wind can transport and accumulate snow unevenly, creating unstable snow slabs.

5. Earthquakes and Vibrations

Ground movement may disturb unstable snow layers.

6. Human Activities

- Skiing
- Trekking
- Construction
- Movement of vehicles or people in unstable areas

Avalanche Formation

Heavy snowfall / wind loading → Weak snow layer → Increasing stress → Loss of stability → Snow slab breaks → Rapid downslope movement

Effects

- Death and injuries
- Burial of people and animals
- Damage to roads and buildings
- Destruction of vegetation
- Disruption of mountain transport
- Temporary blockage of valleys

Mitigation

- Avalanche forecasting
- Hazard-zone mapping
- Controlled release of unstable snow
- Snow fences and barriers
- Restriction of movement in high-risk areas
- Early-warning systems
- Proper planning of mountain infrastructure

GLACIAL LAKE OUTBURST FLOODS (GLOFs)

A **Glacial Lake Outburst Flood (GLOF)** is a sudden and often catastrophic release of water from a glacial lake.

The lake may be retained by a natural dam made of **ice, rock debris or loose moraine material**.

Major Causes

1. Rapid Snow and Ice Melt

Rising temperatures can increase the volume of water stored in glacial lakes.

2. Heavy Rainfall

Intense rainfall can rapidly raise lake-water levels.

3. Avalanches

A large mass of snow or ice entering a lake can generate powerful waves and cause overtopping.

4. Landslides

A landslide entering a lake can displace water and weaken or breach the natural dam.

5. Earthquakes

Earthquakes may destabilise the lake basin or its natural dam.

GLOF Formation

Glacial melting → Lake expansion → Increasing water pressure → Weakening/overtopping of natural dam → Dam failure → Sudden water release → Downstream flood

Why GLOFs Are Dangerous

GLOFs can travel rapidly through narrow mountain valleys and carry:

- Huge volumes of water
- Rocks
- Sediments
- Ice blocks
- Trees and other debris

Therefore, their destructive power can extend far downstream.

Effects

- Loss of life
- Destruction of settlements
- Damage to bridges and roads
- Destruction of hydropower projects
- Agricultural losses
- River-channel modification
- Severe disruption of downstream communities

Mitigation

- Monitoring glacial lakes
- Remote sensing and satellite observation
- Early-warning systems
- Controlled drainage of dangerous lakes
- Strengthening natural dams where feasible
- Hazard mapping
- Evacuation planning
- Downstream disaster preparedness

Important Distinction

GLOF ≠ Ordinary Flood

A GLOF specifically involves the **sudden release of water from a glacial lake**, usually because of failure or overtopping of its natural dam.

DUST STORMS

A **dust storm** occurs when strong winds lift large quantities of loose, dry soil and fine sediment into the atmosphere.

They are particularly common in **arid and semi-arid regions**.

Major Causes

Natural Factors

- Strong winds
- Prolonged drought
- Low rainfall
- Extremely dry soil
- Loose surface sediments
- Extreme weather conditions

Human Factors

- Deforestation
- Overgrazing
- Poor agricultural practices
- Excessive removal of vegetation
- Improper land management

Formation

Drought / low rainfall → Dry and loose soil → Sparse vegetation → Strong winds → Suspension of fine particles → Dust storm

Role of Vegetation

Vegetation acts as a natural protective cover.

Roots bind soil particles and vegetation reduces wind speed near the ground.

Therefore:

More vegetation → Greater soil stability → Less wind erosion

Whereas:

Less vegetation → Exposed soil → Greater wind erosion → More dust storms

Effects

Environmental Effects

- Soil erosion

- Loss of fertile topsoil
- Reduced visibility
- Degradation of agricultural land
- Atmospheric pollution

Human Effects

- Respiratory problems
- Traffic accidents
- Disruption of aviation
- Damage to crops
- Reduced visibility
- Disruption of communication and daily activities

Prevention

- Afforestation
- Shelterbelts
- Windbreaks
- Controlled grazing
- Sustainable farming
- Soil-moisture conservation
- Proper irrigation
- Prevention of deforestation

QUICK COMPARISON TABLE

Disaster	Main Material	Main Trigger	Major Hazard
Landslide	Rock, soil, debris	Rainfall, earthquake, human activity	Downslope mass movement
Avalanche	Snow and ice	Heavy snowfall, warming, vibration	Rapid snow movement
GLOF	Water + debris	Glacial lake dam failure	Sudden downstream flood
Dust Storm	Fine soil and dust	Strong winds + dry conditions	Airborne dust and soil erosion

IMPORTANT CONCEPTUAL LINKS

Rainfall and Landslides

Heavy rainfall → Water infiltration → Increased weight + pore pressure → Reduced stability → Landslide

Climate Warming and GLOFs

Rising temperature → Faster ice/snow melt → Expansion of glacial lakes → Greater pressure on natural dams → Possible GLOF

Temperature and Avalanches

A sudden temperature increase may weaken snow bonding by causing partial melting, thereby increasing avalanche risk.

Vegetation and Dust Storms

Vegetation loss → Exposed soil → Wind erosion → Dust storms

NOTES

Karst Topography:

A distinctive landscape developed mainly in soluble rocks such as limestone through chemical dissolution by groundwater.

Sinkhole:

A depression formed when the ground collapses into an underground cavity or when soluble rock is dissolved.

Landslide:

The downslope movement of soil, rock or debris under the influence of gravity.

Avalanche:

The sudden downslope movement of snow and ice, often mixed with rock and debris, on a mountain slope.

GLOF:

A sudden release of water from a glacial lake caused by overtopping or failure of its natural dam.

Dust Storm:

A weather event in which strong winds lift and transport large quantities of loose, dry soil and fine particles.

Landforms – Formation and Significance

1. Dynamic Nature of the Earth's Surface

The Earth's surface is **constantly changing** because of the interaction of powerful **internal (endogenic)** and **external (exogenic)** forces.

- **Internal forces** originate within the Earth and are responsible for large-scale crustal deformation.
- **External forces** operate on or near the Earth's surface and gradually modify existing landforms.

2. Internal Forces (Endogenic Forces)

Internal forces include:

- **Earthquakes** – sudden movements of the Earth's crust caused mainly by tectonic activity.
- **Volcanic eruptions** – release of magma, gases and ash onto the surface.
- **Folding** – bending of rock layers due to compressional forces.
- **Faulting** – fracturing and displacement of rocks along faults.

Major effects:

These processes can create **mountains, plateaus, valleys and ocean basins.**

3. External Forces (Exogenic Forces)

External forces continuously modify the Earth's surface through:

Weathering → Erosion → Transportation → Deposition

- **Weathering:** Breakdown of rocks in their original position.
- **Erosion:** Removal and wearing away of weathered material.
- **Transportation:** Movement of sediments by water, wind, ice or gravity.
- **Deposition:** Accumulation of transported sediments in a new location.

4. Interaction of Internal and External Forces

Landforms are the result of a **continuous balance between uplift and denudation.**

- Internal forces generally **build and elevate** the Earth's surface.
- External forces generally **wear down, transport and redistribute** materials.

Therefore, the present landscape is the product of long-term **geomorphic processes.**

5. Diversity of Landforms

These processes produce a wide range of landforms, from:

- **Highest mountain peaks**
- **Deep valleys**
- **Plateaus and plains**
- **River valleys and deltas**
- **Coastal features**
- **Deep ocean floors**

6. Importance of Landforms to Human Life

Landforms strongly influence human activities by determining:

- **Climate and local environmental conditions**
- **Availability of natural resources**
- **Agricultural possibilities**
- **Settlement patterns**
- **Transportation routes**
- **Economic activities**
- **Cultural development**

For example, fertile plains generally support dense settlements and intensive agriculture, whereas steep mountainous regions often have limited settlement and transport possibilities.

7. Landforms and Natural Disasters

Understanding landforms is important for **disaster preparedness and risk reduction.** Areas with particular physical characteristics may be more vulnerable to:

- Earthquakes
- Landslides
- Avalanches
- Floods
- Volcanic eruptions
- Glacial lake outburst floods (GLOFs)
- Dust storms